

RIGHT CAPACITORS FOR THE RIGHT MARKET

Japanese capacitor major Nichicon aims to strengthen its foothold in the Indian market with its innovative and market-driven product ranges

Japanese capacitor manufacturer Nichicon is ramping up its presence in India. Having entered the Indian market back in 2012, it has forged alliances with some of the major distributors of capacitors along with setting up its own sales office in India.

Headquartered in Kyoto, Japan, the company has a widespread global presence with 16 manufacturing units, of which 11 are in Japan, four in China and one in Malaysia, and sales offices in the USA, the UK and Asia-Pacific region. The company manufactures aluminium electrolytic capacitors, film capacitors and circuit products, catering to four key markets, namely, automobiles, white goods and industrial inverters, energy, environment and medical, and telecom.

The company has registered gross revenue of US\$ 892 million, with Japan and Asia contributing to as much as 38.4 per cent and 47.7 per cent, respectively. The company claims to have introduced advanced products ahead of the needs of its customers in the market by focusing on technological resources that they specialise in, such as aluminium-etched foil manufacturing technology, electrolyte development and film deposition technology.

Key market segments to cater

Nichicon has identified the capacitor needs of the key markets and will be fulfilling that through the introduction of new products and technology. Here are some of the requirements: *Automobiles and automotive-related equipment:*

- High temperature support (installed in engine rooms)
- Vibration resistance (automotive environments)
- Low temperature ESR characteristics (cold-climate support)

White goods and industrial inverter equipment:

- Miniaturisation (miniaturisation of equipment)
- Charging/discharging support (motor drive circuits)
- Higher withstand voltages (higher efficiency of equipment)

Energy, environment and medical equipment:

- Long life (making wind power generation maintenance-free)



- Higher withstand voltages (higher efficiency of equipment)
- Telecommunications equipment:*
- Thinner (thinning of mobile equipment)
- Lower ESR (instantaneous supply to semiconductors)

Innovative products on offer

The company claims to be thriving on the back of its R&D capabilities and ahead of market innovations. It has been proactive when it comes to introducing products in the market that are linked with the current requirement or demand of consumers. Here are some of the latest innovations that they are offering.

Addressing the high temperature high voltage withstand need of renewable energy sector. As the industrial market makes advances in efficiencies and energy-saving functions, higher voltages are required. This creates a demand for higher voltage aluminium electrolytic capacitors that are used in the control circuits of power supplies. The need is particularly prevalent in renewable energy-related fields such as solar and wind power generation.

Nichicon has met these needs with its LGN series of compact snap-in terminal-type aluminium electrolytic capacitors, which can withstand rated voltages of up to 500V and are guaranteed to operate at 105°C and therefore suitable for the power electronics market, where energy-efficient operations are a priority. Recently, they developed a new electrolyte solution for aluminium electrolytic capacitors that is capable of withstanding voltages of 750V and higher with operating temperatures of 105°C. Primary ap-

plications for this product will include high-voltage converters and will cater to power electronics market, where energy-efficient operations are a priority.

Shaping up ultra-slim capacitors to fit into tablets and mobile devices: With the rapid expansion of tablets and mobile devices, there has been a wide range of peripheral equipment developed that takes advantage of these products' networks. This field is demanding increasingly smaller components that offer more advanced functions. It is also crucial that components use space effectively in order to fit into the shapes of this peripheral equipment.

Nichicon's previous cylindrical-shaped EDLC measured $\phi 6.3 \times 9L$ mm and had the industry's highest capacitance of 1F. A new ultra-slim $\phi 4 \times 30L$ mm product has been developed recently. Despite its thinness, it still has a capacitance of 1F. The EDLC's ultra-thin shape allows it to fit into small spaces and makes it ideal for applications like stylus and other peripherals used with tablets and mobile devices. It can also be used for backup power in wearable technology. And with its ability to be charged on very slight electrical currents, it holds promise for application in the field of energy harvesting.

Introducing ultra-compact capacitors for power and resource efficiency: Due to the increasing needs for power and resource efficiency in industrial machinery, equipment is becoming increasingly smaller. The components also need to be smaller. Nichicon has introduced the industry's smallest class of snap-in terminal-type aluminium electrolytic capacitors till date. Potential applications include inverters for power electronics and switching power supplies. These ultra-compact capacitors are guaranteed to operate at 105°C and are ideal for space-efficient circuit design.

With renewed focus on increasing manufacturing activities in India, especially in the electronics sector, such innovative capacitors will be more and more in demand. All sectors that the company is targeting, be it renewable energy, tablets and mobile devices or industrial power supply, are gathering pace and are looking for such innovative solutions. Perhaps, the time is ripe for Nichicon to ramp up its presence.